Last Updated: May 26, 2026

BatLog v0.1 Single Antenna Instruction Manual

A Low-Cost Modular Passive Integrated Transponder (PIT) Tag Logger for Monitoring of Bat Roosting and Behavior

Purpose: This document provides standardized procedures for the assembly, testing, deployment, and maintenance of a PIT tag logging system using an RFID antenna, microcontroller, SD card data logging, and RTC timekeeping. The SOP is designed to allow replication of the system by external laboratories without prior system familiarity.

Abstract: Automated monitoring of individually marked animals enables the collection of high-resolution activity data, yet commercial systems capable of continuous individual organism tracking remain inaccessible to many research programs. BatLog is a modular, open-source passive integrated transponder (PIT) tag logging system built on Arduino-compatible hardware, costing ~3.5% of average commercial systems. BatLog is designed to extend the application of existing PIT tag infrastructure by incorporating individual identification data into continuous behavioral datasets, without requiring additional animal handling or marking. The system is well-suited for studies conducted in controlled or semi-controlled environments with defined movement points, such as nest boxes, roost entrances, burrows, or other confined passageways. BatLog is most effective in study systems where animal movement is constrained through a defined opening, and individuals reliably pass through the antenna. A dual-antenna configuration enables directional detection, allowing each tracking interaction to be classified as an animal entry or exit event. Deployed in a controlled laboratory setup at Bucknell University, USA, the BatLog system recorded 80,109 paired directional events across 27 individually identified bats. The Arduino-compatible architecture supports integration with additional sensors, including temperature, humidity, accelerometers, and acoustic monitors, making BatLog broadly applicable to behavioral and ecological studies across taxa.

Written By: Angel Yanga (Undergraduate Student, Bucknell University)
Date: November 1, 2024
Last Updated: See the most recent Git commit

Last Updated: May 26, 2026

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1. Overview

This SOP provides a step-by-step procedure to assemble, test, and deploy an RFID Logger system capable of reading PIT tags (134.2 kHz, FDX-B, Biomark HPT9). The system is designed to be modular, low-cost, and portable, including an integrated RTC for timestamping and microSD card data storage.

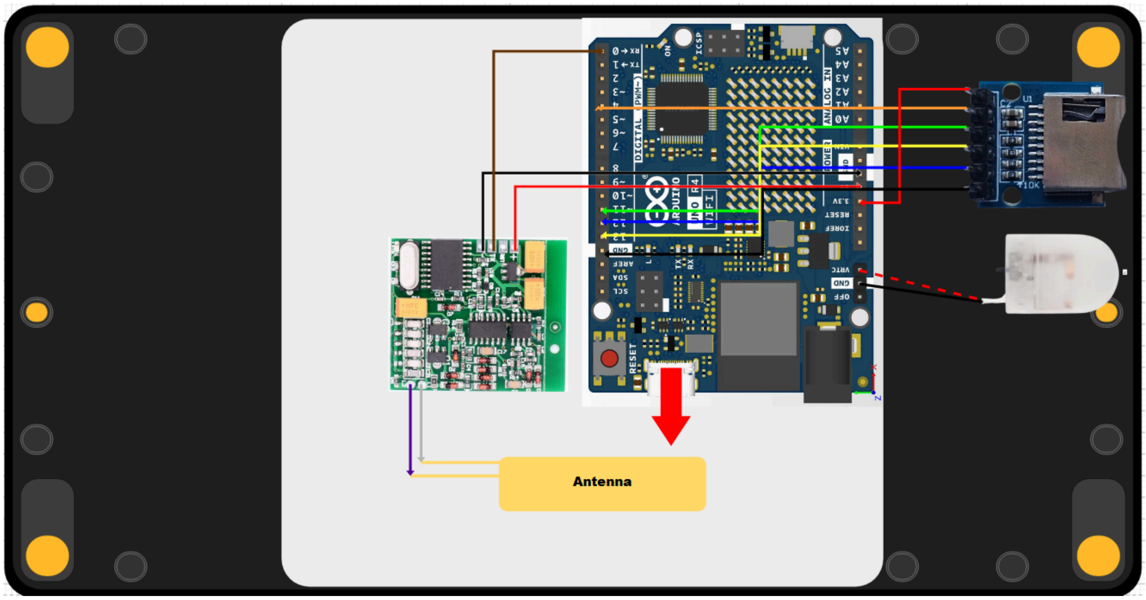


Figure 1. Electronics schematic for the single-antenna PIT tag logging system, showing the Arduino R4 WiFi microcontroller, single RFID-antenna subsystem, SD card module, and power distribution for bat box data collection.

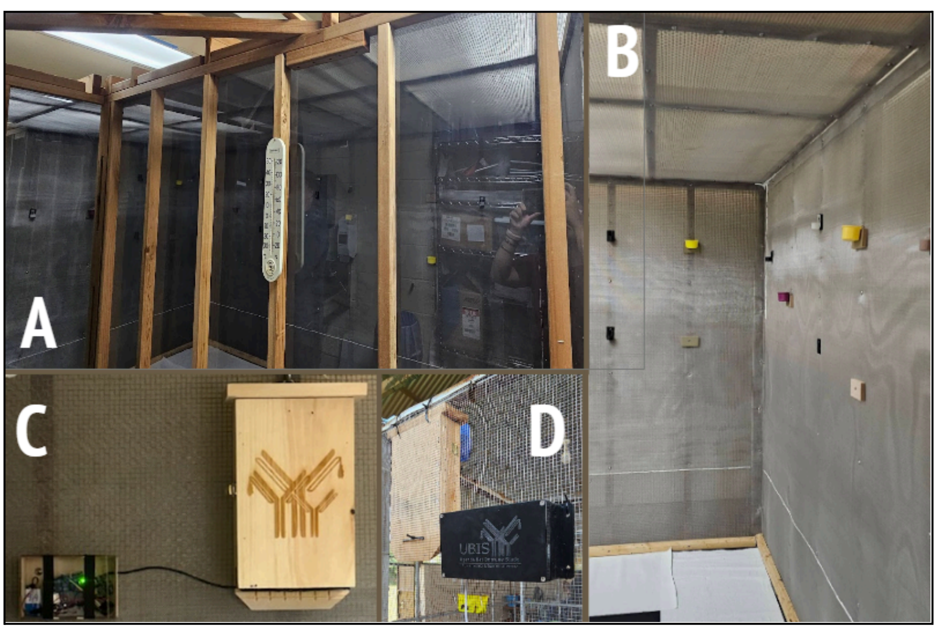


Figure 2. Overview of the flight cage and experimental setups. (A) Exterior view of the flight cage at Bucknell University, showing the acrylic glass panels. (B) Interior view of the flight cage, highlighting the mesh structure. (C) Experimental setup at Bucknell

University, where the system was mounted using wall anchors. (D) Field setup in Uganda, where the wooden bat box was placed inside the enclosure, and the Batlog electronics were mounted externally and secured to the mesh using zip ties.

2. Required Skills

2.1. Electronics Assembly & Hardware

1. **Through-hole Soldering**
   a. Installing pin headers on PCBs
   b. Identifying a functional solder joint
   c. How to solder: Soldering Tutorial for Beginners: Five Easy Steps
2. **Crimping Dupont Connectors**
   a. Properly crimping thin wires to attach dupont connectors
   b. Using a ratcheting crimper correctly
   c. How to crimp for dupont connectors:
   Dupont Connectors - Quickly and easily make your own
3. **Basic Wiring Practices**
   a. Correct power, ground, and signal routing
   b. Follow color-coded wire management
   c. Know the differences and function of different terminal on PCBs (microcontrollers and modules)
   d. Tips for working with dupont wires:
   #12 Five Tricks for working with Dupont wires
4. **Multimeter Use**
   a. Measuring voltage and current to verify functional connections
   b. It's helpful to measure voltage for power connections
   c. It is more useful to measure current between non-power connections (e.g. TX pin to RX pin)
   d. How to use a multimeter:
   How to Use a Multimeter for Beginners - How to Measure Voltage,…

2.2. Microcontroller & Embedded Systems

1. **Arduino IDE Proficiency**
   a. Selecting the correct board and prt
   b. Reliably upload firmware to the system
   c. How to install and run Arduino IDE:
   How to Install Arduino Software IDE on Computer / Laptop
2. **R Studio Proficiency**
   a. Effectively import the raw data, metadata, and produce effective visualizations of findings
   b. How to install and run R Studio:
   How to install R and RStudio on Windows 11 (Updated 2025)
3. **Basic Embedded Debugging**
   a. Interpreting Serial Monitor output

b. Identifying whether failures are firmware vs. hardware

4. RTC Handling

a. Flashing code properly to initialize RTC time

b. Know how to replace the coin battery without losing the stored time

i. You plug in the system to a reliable power source through the Arduino R4’s Type C input.

ii. Then, you replace the coin battery

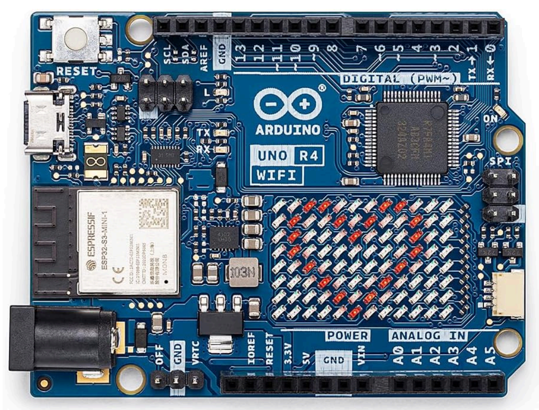
3. Materials

See the full bill of materials in Google Drive for links to materials.

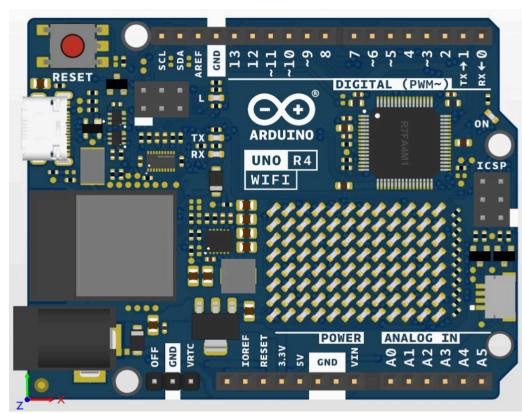
3.1. Electronics

- 1x Arduino R4 WiFi with Integrated Real-time Clock (RTC) Module

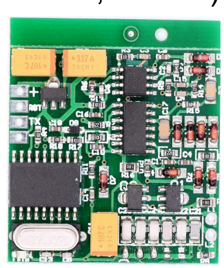
Physical



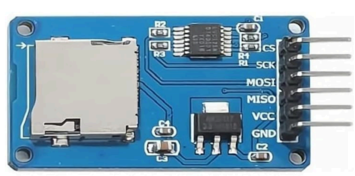
Digital



- 1x RFID Module (Reads 134.2 kHz, FDX-B)



- 1x microSD Card module

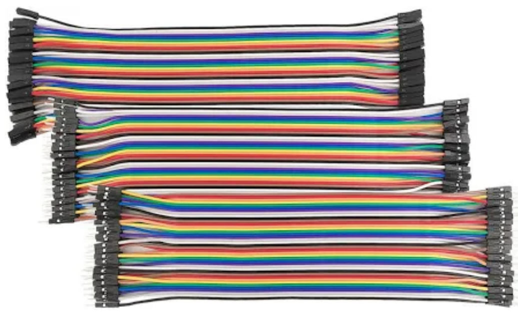


- 1x 2GB microSD Card (Cloudisk)
- 1x 6 1/8 in x 1.375 in x 1/4 in 3D-printed antenna frame
- 1x Roll 24 AWG Copper Wire Antenna

- 1x CR2032 Li-ion Coin Battery
- 1x Battery Holder for CR2032 with pre-attached +/- Wires

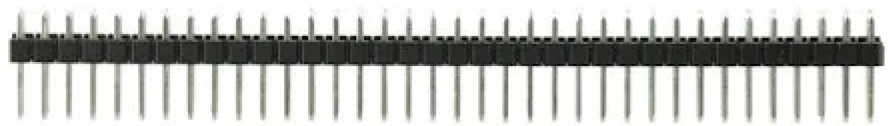
3.2. Wires & Cables

- 1x Type C to USB A Cable, 10 ft, nylon braided
- 1x Dupont Jumper Wire Kit (40 male-to-male, 40 female-to-female, 40 male-to-female, 10 cm, multicolor)



- 1x Dupont Connector Kit, unsoldered, to assemble module connections
  - Needs to include male and female dupont connectors (aluminum part) and headers (plastic part)

- 1x Unsoldered Pin Header Kit
  - Could already be included in dupont connector kit, but not required



3.3. Set Up & Mounting

- 1x Electronics Box; water-resistant, ABS Plastic, includes lid (7.87"x4.72"x2.95")

- 1x Type C to USB A Cable, 10 ft, nylon braided
- 1x Anker Power Bank (PowerCore 10K, 5V/3A USB-C, 10,000 mAh)
- 1x Plastic container without lid
  - Should fit the 5.99 in × 2.81 in × 0.61 in power bank when turned over
- 4x Nylon Zipties (2.5x150 mm)
- 1x Wooden Bat Roost Box
  - *Ours was modified from a commercial bat box kit*

3.4. Tools & Equipment

- 1x Soldering Station with Accessories (Lead, flux, steel wool, lead sucker)

Soldering Station

Accessories

- 1x Wire Stripper & Cutter
- 1x Philips Screwdriver
- 1x Multimeter
- 1x Wire Crimper (0.25-size slot)

- 1x Roll Electrical Tape
- 1x Roll Double-Sided Adhesive

Reference: Uline Removable Double-Sided Foam Squares - 1 x 1", 1⁄16" Thick, but anything similar works.

- 1x MicroSD card reader to USB A/C
- 1x Computer with Arduino IDE and R
- 1x Biomark Test Tag 12mm FDX-B Key Chain

4. System Flow

4.1. Assemble Subassemblies

The following subassemblies may be prepared simultaneously by different personnel to reduce the total build time:

- [Section 4.1] RFID Module
- [Section 4.2] Antenna
- [Section 4.3] SD Card Module
- [Section 4.4] RTC Coin Cell Battery Holder

4.2. Assemble the microcontroller

This will be the main assembly of the system, where you put all of your subassemblies together.

4.3. Test the system

Verify that the assembled microcontroller and all connected subassemblies (RFID module, antenna, SD card module, and RTC battery holder) power on correctly, communicate as expected, and record data without errors.

4.4. Mount the system

Securely install the fully assembled system at the designated deployment location, ensuring all components are properly fastened, protected, and oriented for reliable operation.

4.5. Change the power bank every 3-5 days

Replace the power bank at regular 3-5 day intervals to maintain uninterrupted power and continuous data collection.

4.6. Extract & Process Data

Retrieve the recorded data from the system and clean, organize, and process it for analysis.

4.7. Interpret Results

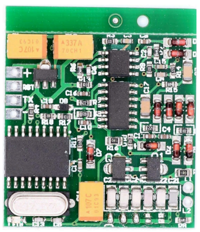
Analyze the processed data to identify patterns, trends, and findings relevant to the study objectives.

5. Detailed Procedure

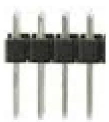
5.1. RFID Module Assembly

Materials & Tools

1. 1x RFID Module (Reads 134.2 kHz, FDX-B)

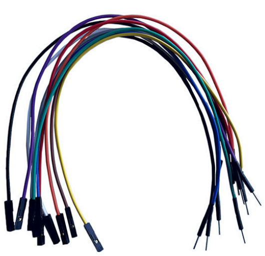


2. 1x 1×4 unsoldered pins
    a. Cut the long strip of unsoldered pin headers into 1x4:



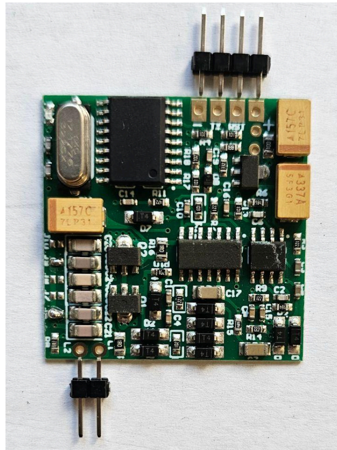
3. 1x 1×2 unsoldered pins
    a. Cut the long strip of unsoldered pin headers into 1x2:

4. 5x Female to male dupont jumper wires
    a. Red, Black, Brown, Purple, Gray



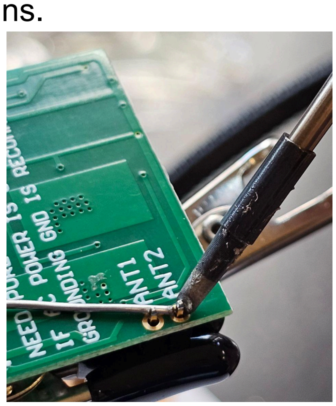
5. 1x Soldering Stationwith Accessories

Procedure



1. Heat the soldering iron to ~310-320°C.
    a. You can heat it to 380°C if needed, as some soldering leads melt at a higher temperature.
2. Insert the dupont 1×2 header into the antenna input holes using the same orientation.

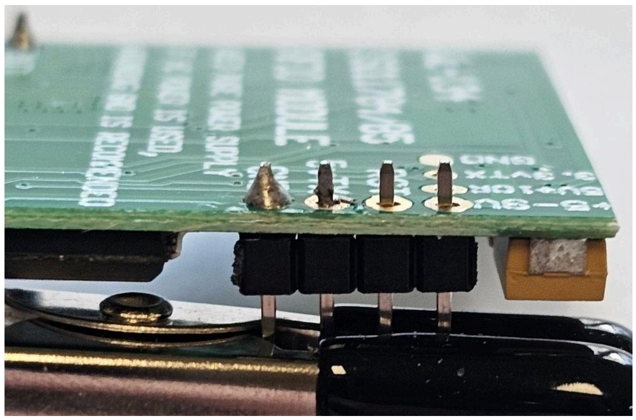
3. Solder both antenna pins.



4. Long pins on top, short pins on bottom.
    a. The long side of the headers will be where your wires can connect, so

make sure you leave them in the most accessible spot for future wiring

5. Solder the two pins securely.
6. Insert the dupont 1×4 header into the RFID module output holes from the back.
    a. Output holes will be recognized by:
        i. TX
        ii. RX (won't be used, but it's convenient to solder)
        iii. 5-9+V (or 3.3V)
        iv. GND



7. Solder all four pins securely.
8. Attach female dupont wires using the following color convention:
    a. +5–9 V → Red
    b. TX → Brown
    c. GND → Black
    d. ANT1 → Purple
    e. ANT2 → Gray

5.2. Antenna Assembly

Materials & Tools

1. 1x 6 1/8 in x 1.375 in x 1/4 in 3D-printed antenna frame
2. 1x Roll 24 AWG Copper Wire Antenna
3. 1x Roll Electrical Tape

Procedure

1. Wrap the 24 AWG copper wire on one side of your mold with about 5 inches of spare sticking out
    a. You need the space copper wire to connect to an RFID Module, so this is very important
2. Wrap the rest of the 24 AWG wire around the antenna mold approximately 40 turns.
    a. Avoid crossing turns excessively to make sure it is as flat as possible after each wrap.
3. At the end, leave the same length of copper wire sticking out as in step 1.
    a. This is your second antenna lead to connect to the RFID Module

4. Carefully use electrical tape to cover exposed wires.
5. Progress outward slowly, ensuring the wire does not unravel.
6. Fully secure the antenna with electrical tape once wrapping is complete.

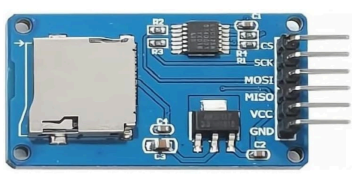
Notes

- I do not have sufficient images for this process step-by-step. So, you can use this video (https://youtu.be/lRoxlW0Hx-g: 8:37-12:40) as a visualization.
  - After wrapping the antenna around the PVC, we would wrap our antenna with electrical tape. This is for protection for and form the bats.
- This was the most sensitive step for our production of 10 BatLogs for the Moco 2025 field study. This is due to the size of the copper wire we used. It was too thin to crimp, so we needed to get each crimp perfectly.
  - If you plan on tracking larger animals, thicker copper wire would be ideal and you might not have this problem.
- If the crimping is giving you a difficult time, a transition to soldered JST connectors could be helpful.
  - We soldered the antenna leads (the excess antenna wire) to a male JST connector. Then, the antenna leads on the RFID modules were soldered with a female-ended JST connector.
  - The difficult part here would be soldering very tiny parts together, but this method was more secure than crimping.

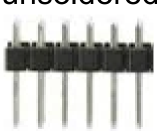
5.3. SD Card Module

Materials and Tools

1. 1x microSD Card module



2. 1x 1×6 unsoldered pins



3. 6x Female to Male dupont jumper wires
   a. Red, Orange, Yellow, Green, Blue, Black
4. 1x 2GB microSD Card (Cloudisk)
5. 1x Soldering Station with Accessories

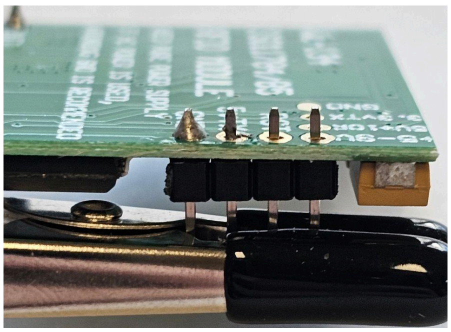
Procedure

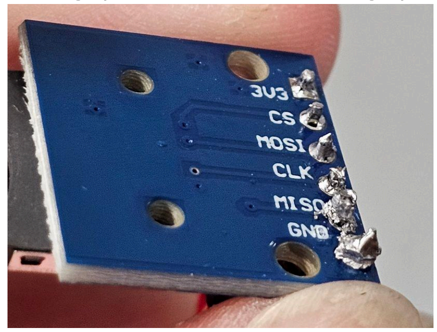
** If your SD Card Module comes with pre-soldered pins, you can skip steps 2 through 5*

1. Heat the soldering iron to ~310-320°C.
   a. You can heat it to 380°C if needed, as some soldering leads have a

higher melting point.

2. Insert the 1×6 unsoldered pins into the SD card module from the back side of the PCB.
    a. The long pins must extend through the top of the module.
    b. The short pins remain on the back side of the PCB.
3. Verify correct orientation before soldering (see reference image).





4. Solder each header pin to its corresponding contact hole on the SD card module.
    a. Heat both the pin and the pad simultaneously before applying solder.
    b. Ensure the solder fully wets both surfaces with no visible gaps between the pin and the pad.
    c. A proper solder joint should resemble an inverted cone extending from the pad to the pin.
5. Visually inspect all solder joints and gently tug each pin to confirm stability.
6. Attach female dupont jumper wires to the header pins using the following color-to-pin mapping:
    a. 3V3 → Red
    b. CS → Orange
    c. MOSI → Green
    d. CLK/SCK → Yellow
    e. MISO → Blue
    f. GND → Black
7. Insert the microSD card into the SD card module.
8. Ensure the card is oriented correctly and fully seated in the module’s slot according to its orientation.

5.4. RTC Coin Cell Battery Holder

Materials and Tools

1. 1x Battery Holder for CR2032 with pre-attached +/- Wires
2. 2x Female dupont connectors (aluminum)
3. 1x Wire Cutter
4. 1x Wire Crimper (0.25-size slot)
5. 1x Pliers

Procedure

1. Using a wire cutter, separate two female connectors from the connector strip.
2. Hold the end of one battery holder wire firmly to prevent movement during

positioning.

3. Position the wire within the female crimp terminal as follows:
    a. Place the insulated portion of the wire between the triangular tabs of the terminal.
    b. Place the exposed conductor between the rectangular tabs of the terminal.
4. While holding the wire and terminal steady with one hand, use pliers to gently pre-close the tabs:
    a. Fold the rectangular tabs over the exposed conductor.
    b. Fold the triangular tabs over the wire insulation.
    c. Ensure only the tabs are bent; the rest of the terminal body must remain unaltered.
5. Align the partially closed terminal and wire with the 0.25-size crimping slot on the crimping tool.
    a. Confirm that only the tabs are inside the crimping jaws.
    b. Avoid placing any other portion of the terminal body into the crimper.
6. Crimp the terminal by fully pressing the crimping tool until it releases automatically.
    a. If the tool does not release, apply additional pressure until it opens.
    b. Do not pull the wire while the crimper is engaged.
7. Repeat Steps 2–6 for the second battery holder wire.

Notes

- The shorthand of this procedure is taking a non-crimped +/- end and crimping it with dupont connectors.

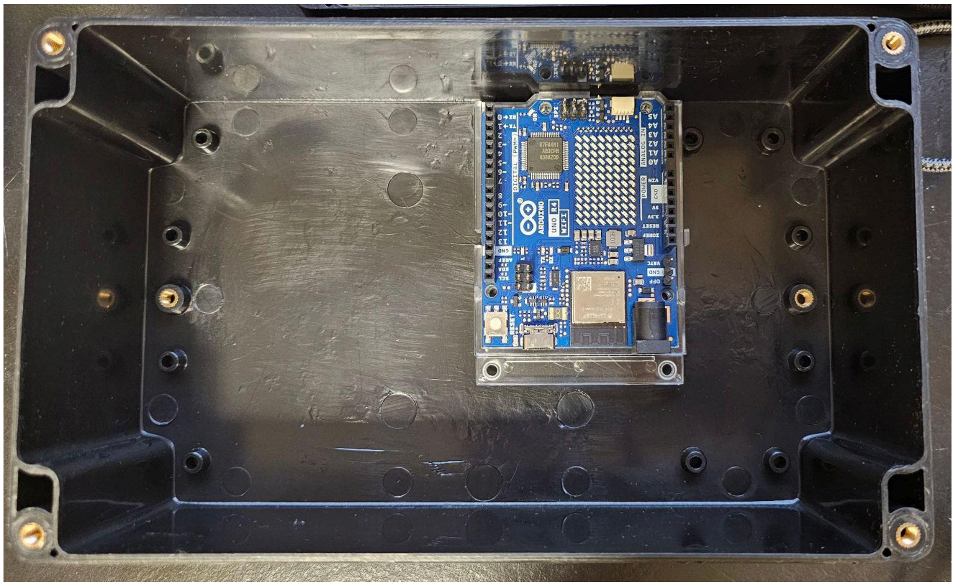
5.5. Microcontroller Setup (Arduino R4 Wi-Fi)

Materials

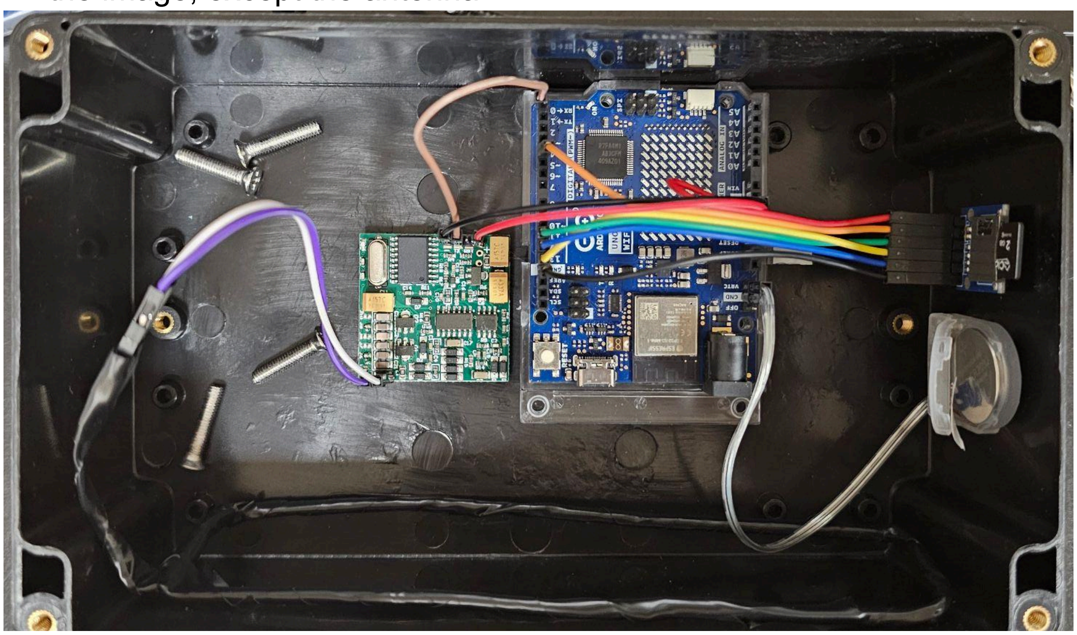
1. 1x Arduino R4 WiFi with Integrated Real-time Clock (RTC) Module
2. 1x RFID Module Subassembly
3. 1x Antenna Subassembly
4. 1x SD Card Subassembly
5. 1x Electronics Box; water-resistant, ABS Plastic, includes lid (7.87"x4.72"x2.95")
6. 1x Type C to USB A Cable, 10 ft, nylon braided
7. 1x Computer with Arduino IDE and Arduino Script
8. 1x Roll Double-Sided Adhesive

Procedure

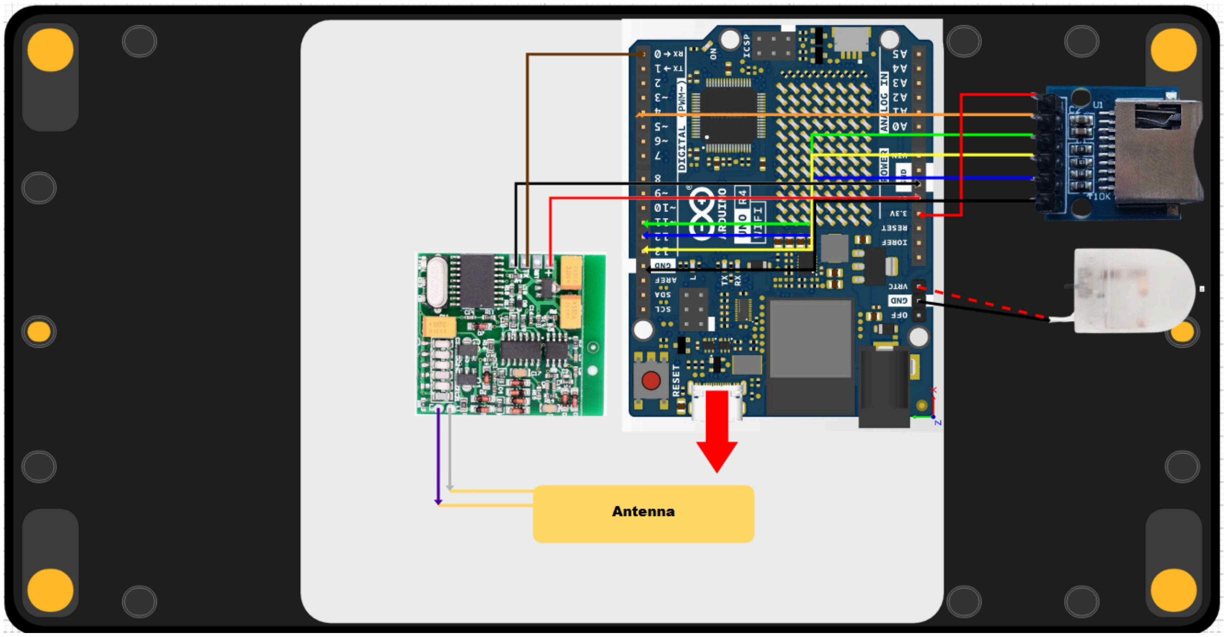
1. The Arduino R4 Wi-Fi is ready for assembly right out of the box.
    a. It is recommended to test the board with pre-installed test scripts on the Arduino IDE
2. Mount the Arduino Board in the electronics box with double-sided adhesive.



3. Mount the rest of the subassemblies with double-sided adhesive according to the image, except the antenna.



4. Use all of the dupont wires from the subassemblies to connect to the Arduino
   a. Follow the diagram below:



5. Connect the antenna with the gray and purple wires of the RFID Module
6. Connect the system to your computer with the Arduino IDE with the charging cable
    a. A cable hole through your electronics box is necessary, so you can have the charging cable run through it once the lid is closed
7. Open the Arduino Script file (ino file) on Arduino IDE
8. Uncomment line 78 by deleting the "//"
    a. The line of code should change colors to non-gray text

Commented:

Arduino_Single_Antenna_Ver_1.0_11.11.2024.ino

```
76    RTCTime startTime(dayofMon, convertMonth(month) , year, hours, minutes, seconds,
77                              convertDOW(dayofWeek), SaveLight::SAVING_TIME_ACTIVE);
78    //RTC.setTime(startTime); //COMMENT AND UNCOMMENT THIS LINE
79  }
```

Uncommented:

Arduino_Single_Antenna_Ver_1.0_11.11.2024.ino

```
76    RTCTime startTime(dayofMon, convertMonth(month) , year, hours, minutes, seconds,
77                              convertDOW(dayofWeek), SaveLight::SAVING_TIME_ACTIVE);
78    RTC.setTime(startTime); //COMMENT AND UNCOMMENT THIS LINE
79  }
```

9. Upload the code to the Arduino R4 WiFi Board selected
10. Comment line 78 by typing "//"
11. Reupload the code to the Arduino R4 WiFi Board selected

Steps 9-11 are critical to ensure accurate time capture on the internal RTC Module of the Arduino

12. Open the Serial Monitor on Arduino IDE (Ctrl+Shift+M)
13. Press the white reset button on the Arduino

14. You will see messages on the Serial Monitor
    a. If your wiring is functional, you should see this message:

```
Starting Serial Communication...
Ready to read RFID...
Starting RTC Module and Setting Time
```

15. Put the Biomark HPT9 Tag tester through the antenna
    a. When your system is functional, here is the output on the Serial Monitor:

```
Timestamp: 19/12/2025-14:37:08, RFID Serial Number: 0FA02E7700DD30010000000000r�
```

    b. Here are example outputs when your system has issues:
        i. SD Card not present or undetected

```
Starting Serial Communication...
Card failed or not present.
```

        ii. SD Card works, but the CSV file can't open

```
Starting Serial Communication...
Ready to read RFID...
Error opening RFID_Log.csv
Starting RTC Module and Setting Time
```

        iii. No RFID scanned

```
Starting Serial Communication...
Ready to read RFID...
Starting RTC Module and Setting Time
```

        iv. RFID scanned but not written into CSV

```
Timestamp: 13/10/2024-15:27:20, RFID Serial Number: 0FA02E7700DD30010000000000r�
Error opening RFID_Log.csv for writing
```

16. You are ready to mount the system once it is functional. If not, go through the troubleshooting section.
17. Once you have troubleshooted your errors, you can mount your system.

5.6. Mounting & Deployment

Materials

1. 1x Electronics Box; water-resistant, ABS Plastic, includes lid (7.87"x4.72"x2.95")
    a. 4x included Philips screws for the lid
2. System electronics (fully assembled and tested)
3. 4x Nylon Zipties (2.5x150 mm)
4. 1x Wooden Bat Roost Box
5. 1x Anker Power Bank (PowerCore 10K, 5V/3A USB-C, 10,000 mAh)
6. 1x Type C to USB A Cable, 10 ft, nylon braided
7. 1x Plastic container without lid
    a. Should fit the 5.99 in × 2.81 in × 0.61 in power bank when turned over
8. 1x 2GB microSD Card (Cloudisk)

9. 1x MicroSD card reader to USB A/C
10. 1x Computer for file backup

Procedure

1. Unplug the system from the computer.
2. Unplug the antenna from the system. It will be reattached once mounting is done.
3. Secure the enclosure lid onto the electronics box using four screws, ensuring the charging cable remains routed through the designated opening in the box.
4. Align the electronics box behind the wooden roost box.
5. Use zip ties to mount the electronics box to the outside of the flight cage, positioned near or directly behind the wooden roost box.
    a. The system must be close enough to the roost box opening for the antenna to reach it from behind.
6. Open the wooden bat roost box.
7. Mount the inner antenna about 2 inches inside the box.
    a. Use electrical tape to secure the antenna.
    b. Leave the two antenna leads sticking out through the entrance of the box. These leads must be able to pass through the cage mesh and connect to the RFID module mounted outside the cage.
8. Mount the outer antenna directly at the entrance of the wooden roost box.
    a. Use electrical tape to secure the antenna.
    b. Make sure there are no gaps between the wooden entrance and the antenna so the bats cannot pull it off.
9. Connect the antenna leads to the RFID modules.
    a. If needed, poke holes in the cage mesh to allow the DuPont leads from both antennas to pass through to the electronics system outside the cage.
    b. The inner antenna connects to the top (inner) RFID module, and the outer antenna connects to the bottom (outer) RFID module. Follow the image below to ensure proper connections.
10. Connect the system to the power bank using the USB Type-C cable.
11. Place the power bank on top of the flight cage and cover it with a plastic container to protect it from water exposure.
12. Replace the power bank every 3–5 days.
    a. Although the system can theoretically operate for up to 5 days, field testing showed that replacing the power bank every 3 days leaves approximately 20% charge remaining, providing a safety margin.
13. To check the collected data:
    a. Unscrew the enclosure lid.
    b. Remove the microSD card from the slot.
    c. Insert the microSD card into a computer and open the CSV file.
14. Verify that the CSV file contains accurate timestamps and valid PIT tag readings.
    a. If both are correct, the system is functioning properly.
15. Reinsert the microSD card into the system and reseal the enclosure lid.
16. Do not re-upload the firmware if the system is functioning correctly.

a. After reinserting the microSD card, simply press the reset button on the microcontroller and leave the system running.

17. Once you have completed the experiment, perform the data extraction process (Section 7) before unmounting the system for cleanup.
18. You must cut the zipties off the mesh wall to unmount

Notes

- I do not have sufficient photos of the mounting process. I attached images in the Google Folder (link) for your reference to see how the system was mounted.

6. Troubleshooting

6.1. (1) SD Card not present or undetected, or (2) SD Card works, but the CSV file can't open

Sample Serial Output

Output 1	Output 2
Starting Serial Communication... Card failed or not present.	Starting Serial Communication... Ready to read RFID... Error opening RFID_Log.csv Starting RTC Module and Setting Time

Potential Issues & Fixes

1. The SD Card wires are wired incorrectly
   a. Try to rewire the jumper cables between the SD card subassembly and Arduino
   b. Unplug all the wires on the Arduino from the SD Card subassembly
   c. Unplug the wires on the SD Card Assembly
   d. Follow Procedure Step 6 from Section 4.3 SD Card Module
   e. Follow Procedure Step 4 from Section 4.5 Microcontroller Setup (Arduino R4 Wi-Fi)
   f. Re-run the system and try to read your tester again
2. Soldering pins were not properly soldered
   a. Unmount the SD card subassembly from the electronics box
   b. Resolder the connections by following Procedure Step 4 from Section 4.3 SD Card Module
   c. Re-run the system and try to read your tester again
3. The wiring is functional, but the SD card is not being read
   a. Take out the microSD card by sliding it out
   b. Wipe the microSD card leads with a microfiber cloth
   c. Insert the microSD card into the module securely
   d. Re-run the system and try to read your tester again
4. The wiring is functional, but the SD card isn't all the way in
   a. Remove and reinstall the microSD card
   b. Re-run the system and try to read your tester again

6.2. No RFID Scanned

Sample Serial Output

```
Starting Serial Communication...
Ready to read RFID...
Starting RTC Module and Setting Time
```

Potential Issues & Fixes

1. The antenna is not connected properly
   a. Unplug and replug the dupont connections
2. The antenna dupont heads are not properly crimped
   a. You can re-crimp the ends again to secure the female head to it properly
3. The antenna is not coiled sufficiently
   a. You need to replace another antenna
   b. You need to maintain a strong tight connection between each wire
   c. You need to make sure you wrap it thick enough. You can loop the wire up to 60-70 times.
4. The antenna coil loop is broken
   a. The coil loop can be broken with rigid corners, eventually cutting the wires in half
   b. You need to replace another antenna

6.3. RFID scanned but not written to CSV

Sample Serial Output

```
Timestamp: 13/10/2024-15:27:20, RFID Serial Number: 0FA02E7700DD30010000000000r�
Error opening RFID_Log.csv for writing
```

Potential Issues & Fixes

1. The SD card is the issue here, so follow the troubleshooting procedure on 5.1

7. Data Processing

7.1. How to extract raw data

Materials

1. 1x Electronics system
2. 1x MicroSD card reader to USB A/C
3. 1x Philips Screwdriver
4. 1x Computer for file backup
   a. You can also use your phone in the field

Procedure

1. Unscrew the Phillips screws that secure the lid.
2. Remove the lid of the system.
3. Remove the SD card from the module by sliding it out.
4. Insert the SD card into the card reader.
5. Plug the card reader into your computer.
6. Save the CSV file containing the raw data locally for later data processing.

7.2. How to convert raw data into Animal ID

Our RFID module reads input from the RFID tags in reversed decimal (base-12) order. In the lab, animal IDs are defined as a combination of the project leader's initials and the last five digits of the PIT tag's hexadecimal code. As a result, the raw PIT tag data must be further processed. I used a Google Sheet to perform this conversion.

Procedure

1. Import the raw data from the CSV file into a new tab.
    a. Use Ctrl+A and Ctrl+C to copy the contents of the raw CSV;
    b. Then, Ctrl+V to paste everything into the Raw_Data tab.
2. Copy the column headers from Columns C–G into Row 1 of the Raw_Data tab.
3. Copy the formulas from Columns C–G into Row 2 of the Raw_Data tab.
4. Fill down the formulas, so they apply to all rows of raw data.
5. Export the newly processed Raw_Data tab as a CSV file for later processing in R.

7.3. Processing Data with R

Procedure

1. Create a new project folder.
2. Download "Moco2025PITtagExploration.Rmd" from the Google Drive link.
3. Download your experiment metadata and raw data from the loggers.
4. Move the files downloaded in Steps 2 and 3 into the new project folder created in Step 1.
5. Open the .Rmd file in R.
6. Run the code blocks from top to bottom.
7. Starting at line 226, you can customize how you want to use the combined datasets.

8. Resources

File resources are attached in the shared BatLog Google Drive folder

8.1. Arduino Code

See Google Folder Link & Annotations in the code

8.2. R Code

See Google Folder Link & Annotations in the code

8.3. CAD Files for 3D Prints

See Google Folder Link

8.4. Diagrams & Photos

See Google Folder Link

8.5. Supplementary

- How to Build an Animal RFID reader with Arduino: #263 How to build an Animal RFID reader with Arduino, ESP8266, or ES…
  - This helps you understand the reason why/how the components work together.
- How to solder: Soldering Tutorial for Beginners: Five Easy Steps
- How to install and run Arduino IDE: How to Install Arduino Software IDE on Computer / Laptop
- How to install and run R Studio: How to install R and RStudio on Windows 11 (Updated 2025)
- How to use Onshape: Onshape Tutorial for Absolute Beginners!
  - Onshape is a web-based 3D modeling platform, where you can make your 3D model for your antenna housing.
  - An alternative way of coiling your antenna could be using a rigid PVC pipe (circular) or a short hollow wooden bar (rectangular). You may need a router or table saw to cut down the original material.